

A mathematical modelling of rheumatoid arthritis

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Abstract

We developed a mathematical model to describe the inflammatory dynamics during arthritis development and resolution, focusing on the interaction between three key immune cell populations within the joint, namely neutrophils (N), synovial sublining tissue-resident macrophages (SL-TRM, T), and circulating monocytes that are recruited into the inflamed joint and differentiate into monocyte-derived macrophages (MoMac, M). An inflammatory signal (S), representing the response to an external arthritogenic stimulus, drives the recruitment and activation of neutrophils and Momac, while SL-TRM exert a regulatory and protective role. We propose a minimal ordinary differential equation model of acute inflammatory response into a joint that couples an inflammatory signal S to the three cell populations (N, T, M) and incorporates recruitment, stimulation, nonlinear self-limitation and mutual regulation. We showed that the equilibrium structure depends on three adimensional parameters A, B, C and derived a cubic equation characterizing nontrivial steady states. We performed an analytical study of equilibria and their stability, and identified a bifurcation at $A = 1$ separating a disease-free regime from a sustained inflammatory state. Numerical simulations illustrate typical transient dynamics under various scenarios and show the ability of the model to capture the transition from acute to chronic inflammation, as well as the resolution of inflammation under certain conditions. In conclusion, our model provides a framework to explore mechanisms underlying acute versus sustained joint inflammation and offers a basis for parameter estimation from experimental data, and future extensions toward evaluation of therapeutic strategies.

Keywords : Arthritis; Bifurcation Analysis; Inflammation; Persistence.

1 Introduction

Mathematical modelling of auto-immune disease is relatively recent. According to the systematic survey on mathematical modelling of auto-immune disease by Ugolkov et. al. [14], the first papers date back to the 1970s. Since then, the number of relevant articles increased exponentially, reaching more than 200 publications per year by 2024. However

according to Macfarlane et. al. [12] which provides another careful and exhaustive bibliography in 2020, the number of mathematical models of Rheumatoid Arthritis (RA) is still relatively small. An illuminative survey paper introducing the issues and pitfalls of mathematical modelling in immunology is Andrew et. al. [3]. We refer to Handel et. al. [11] for examples and guidelines for sound mathematical modelling in immunology and also for the free software R-package DSAIRM [8] which allows to easily simulate and analyze a large variety of mathematical models of immune response. Among mathematical models that deal with aspects of RA, we can mention the following ones. In Seymour et. al. [1] a 4 equations model is introduced to describe the dynamics of the population of monocytes and the concentrations of three cytokines, namely IL-6, TNF- α and IL-10. In Jit et. al. [2] RA vs. systemic inflammatory response syndrom is modelled by a system of four differential equations, describing the evolution of population of certain cell compartments, namely TNF- α (free or bound), antibodies and antibody-antigen complexes. The effect of TNF- α inhibitors like (s)TNFR2, Etanercept, Infliximab are investigated. Baker et al.[4] introduced a model with pro-inflammation (p) and anti-inflammation cells (a) and considered a system of two differential equations for p and a. More complex models exhibiting various types of bifurcations are introduced in [6]. In Odisharia et. al. [10] a 5 differential equations system is investigated to simulate the effect of tocilizumab on the populations of B and T-lymphocytes and cell population of cartilage. There exist also spatialized models, see Friedman et. al. [9] where PDE (partial differential equations) are used to model a joint (synovial fluid and cartilage) and Macfarlane et. al. [13] where an hybrid model is introduced to describe pannus formation in a joint. Effects of drugs such as infliximab, which blocks TNF- α ; methotrexate which increases apoptosis of macrophages, and tocilizumab which blocks the receptor of IL-6 are investigated. In Relouw et. al. [15] a 15 equations model with 48 parameters is introduced to model sepsis. In Zhou et. al. [7] a 4 differential equations model is introduced to describe the dynamics of the population of macrophages and fibroblasts cells and their mutual growth factor. A paper that particularly inspired us is Ciuperca et. al. [16] which focus on Alzheimer disease and presents a minimal compartment model with 5 equations describing the concentrations of A β -oligomers, oligomers in the amyloid plaques, A β -monomers, microglial cells, and interleukins.

Here we chose a compromise between simplicity and accuracy to catch essential features. We chose to focus on cell populations and their interactions instead of the fine description of the many cytokines dynamics involved. Specifically, our model, presented in the next section, has 4 equations and only 7 parameters and is an ordinary differential equations systems. The behavior of the system will be shown to depend on a single nondimensional parameter. Despite its simplicity (or due to) we hope that it can bring some understanding of the arthritis process and new ideas for therapy.

The paper is organized as follows. In Section 2 we present the model discussed in the paper. In Section 3 we analyse the mathematical properties of the differential system, compute its steady states and discuss their stability. In Section 4, we present some numerical simulations pertaining to different scenarii. The paper ends with Section 5 concluding remarks and perspectives.

2 Biological Model

In the experimental mouse model of arthritis used in this study, namely Serum Transfer Arthritis (STA), the acute inflammatory response is triggered by an external signal (injection of an arthritogenic serum) that activates the immune system and initiates an inflammatory response within the joint tissue, which is composed of a lining and sub-lining membrane. At steady state, the synovial lining membrane is populated by specialized tissue-resident macrophages (L-TRM) that form a protective barrier and contribute to immune regulation. Tissue-resident macrophages (TRM) are also present within the sub-lining membrane (SL-TRM), scattered and fewer in number. Both TRM contribute to early immune sensing of environmental cues, exerting an overall protective and regulatory role by limiting excessive inflammation. However, during arthritis inflammatory flares, while L-TRM remain preserved, SL-TRM are progressively depleted. This depletion is concomitant with the accumulation of macrophages derived from blood monocytes infiltrating the inflamed joints (Momac). The loss of the SL-TRM population weakens synovial immune regulation and favors persistent inflammatory cell infiltration through to the local production of cytokines and chemokines. This activation promotes the rapid recruitment of neutrophils (N) from the bloodstream into the joint, where they become the dominant cell population during the acute phase of inflammation. Neutrophils amplify the inflammatory response through the release of pro-inflammatory mediators. Based on these biological observations [17], we modeled the joint inflammatory response as the interaction between an inflammatory signal and three cell populations—N, SL-TRM, and Momac—capturing recruitment, amplification, and regulatory mechanisms in a minimal dynamical framework.

2.1 Mathematical Modelling

The variable $S(t)$ is the inflammatory signal at time t . We take as unity the standard dose of the arthritogenic serum, so that $S(t)$ is dimensionless and represents the relative strength of the inflammatory signal with respect to the standard dose. We rescale all variables by their respective baseline values in the disease-free state, so that they are dimensionless and represent relative populations with respect to their baseline levels. Namely variable $N(t)$ (resp. $M(t)$, $T(t)$) designates the relative population of neutrophils (resp. Momac cells, SL-TRM cells) at time t with respect to the baseline population $\hat{N}$ (resp. $\hat{M}$, $\hat{T}$).

The model is given by the following system of ordinary differential equations:

$$\frac{dS}{dt} = (a N - b T) S \quad (1.1)$$

$$\frac{dN}{dt} = \alpha S N - \delta_N N (N - 1) \quad (1.2)$$

$$\frac{dM}{dt} = \beta S - \delta_M M (M - 1) \quad (1.3)$$

$$\frac{dT}{dt} = -\gamma S T + \delta_M M (1 - T) \quad (1.4)$$

where $a, b, \alpha, \beta, \gamma, \delta_N, \delta_M$ are positive parameters.

Let us explain the meaning of each term in the system (1). The growth rate of the inflammatory signal $S(t)$ is enhanced by neutrophils N (factor a) and inhibited by

SL-TRM macrophages T (factor b), which translates into the differential equation (1.1). Neutrophils N are recruited by the inflammatory signal S (factor α) and they have a nonlinear logistic self-limitation (factor δ_N). This is modeled by the term $\delta_N N(N - 1)$ in the equation for N , which is negative when $N > 1$ and positive when $N < 1$, hence in the absence of inflammation it tends to stabilize the population of neutrophils around its baseline level. This is the meaning of equation (1.2).

Equation (1.3) describes the dynamics of the Momac cells M . They are recruited from the blood stream at a rate βS proportional to the inflammatory signal. They have a logistic self-regulation mechanism analogous to that of neutrophils. Note that the linear rate βS of recruitment of Momac cells is not proportional to the current population of Momac cells, which models the fact that the recruitment of Momac cells from the blood stream is not directly influenced by the current population of Momac cells in the joint, but rather by the level of inflammation.

Equation (1.4) describes the dynamics of the SL-TRM cells T . The SL-TRM cells T are destroyed by the inflammatory signal at a rate γS . When T is less than 1, the term $\delta_M M(1 - T)$ is positive and contributes to the growth of T , while when T is greater than 1, this term becomes negative and contributes to the decrease of T . This models the fact that the population of SL-TRM cells is regulated by the population of Momac cells, which can promote their proliferation when they are depleted, but can also inhibit them when they are in excess.

In the system (1), we see that inflammation further increases neutrophils (factor α), increases Momac (factor β), and decreases SL-TRM (factor γ).

Notice that we do not keep track of the population of fibroblasts, but rather the population of neutrophils, which are recruited by inflammatory chemokines.

Remark 2.1. *The term $\delta_M M(1 - T)$ in the equation for T models somehow the stimulation of SL-TRM cells by Momac cells, which is supported by experimental evidence [17]. Note that we could have taken another parameter $\delta_T \neq \delta_M$ in this equation for T , but we choose to keep it simple by limiting the number of parameters, since analysis and numerical simulations show that it does not change the dynamics of the crisis. To make things easier to understand, we summarize the model in the following diagram (see figure 2.1).*

3 Analysis of the model

3.1 Existence and non-negativity of the solution.

Existence and uniqueness of a local solution of the system (1) is guaranteed by the Cauchy-Lipschitz theorem since the right-hand side of the system is locally Lipschitz continuous.

Proposition 3.1. *If the initial conditions $S(0)$, $N(0)$, $M(0)$, and $T(0)$ are nonnegative, then the solution $(S(t), N(t), M(t), T(t))$ of the system (1) remains nonnegative for all $t \geq 0$.*

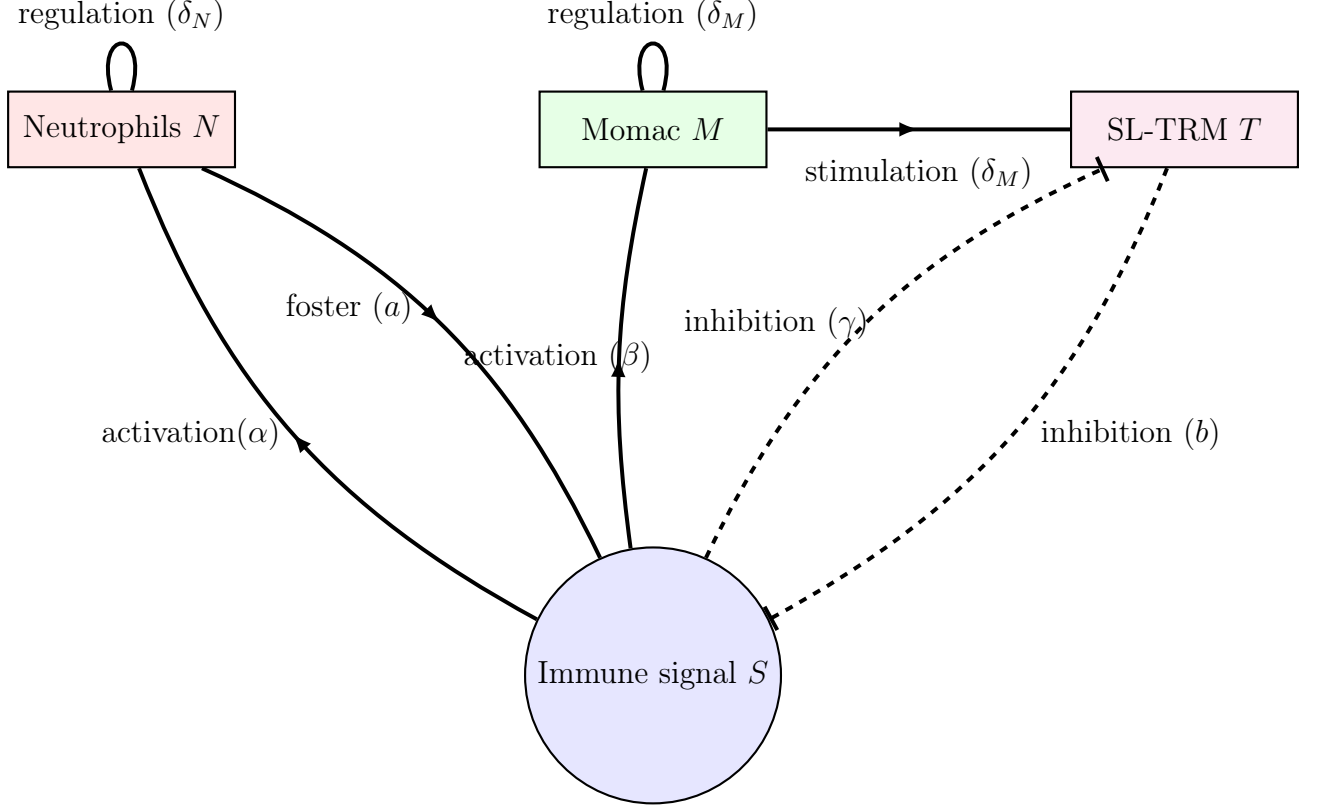


Figure 1: Schematic representation of the model

Proof. We consider the vector field $F = (f_1, f_2, f_3, f_4)$ for $y = (y_1, y_2, y_3, y_4) \in \mathbb{R}^4$ given by:

$$\begin{cases} f_1(y) = (ay_2 - by_4)y_1, \\ f_2(y) = \alpha y_1 y_2 - \delta_N y_2 (y_2 - 1), \\ f_3(y) = \beta y_1 - \delta_M y_3 (y_3 - 1), \\ f_4(y) = -\gamma y_1 y_4 + \delta_M y_3 (1 - y_4). \end{cases}$$

We observe that the vector field F is *quasi-positive*, i.e. for each $i = 1, 2, 3, 4$, we have $f_i(y) \geq 0$ whenever $y_i = 0$ and $y_j \geq 0$ for all $j \neq i$. Thus from Proposition 2.1 in [5], we deduce that if the initial conditions are nonnegative, then the solution remains nonnegative for all $t \geq 0$. $\square$

This property is biologically meaningful since the variables S , N , M , and T represent cell populations, which cannot be negative. Our model thus ensures that the solutions remain within the biologically relevant domain.

3.2 Boundedness of the solution.

Considering the equation for T , one can easily prove that $T(t)$ is always bounded by its initial value $T(0)$ or by 1 if $T(0) \leq 1$, for all $t \geq 0$.

Remark 3.1. *This is biologically relevant since the population of SL-TRM cells is limited by spatial constraints within the joint tissue and regulatory mechanisms that control their proliferation.*

is it biological fact that SL-TRM is bounded by disease-free value?

Proposition 3.2. *Assume that the initial condition $T(0)$ is less than or equal to 1, then the variable $T(t)$ remains less than or equal to 1 for all $t \geq 0$. If $T(0) > 1$, then $T(t)$ remains less than or equal to $T(0)$ for all $t \geq 0$.*

Proof. Let us consider the equation for T :

$$\frac{dT}{dt} = -\gamma ST + \delta_M M(1 - T)$$

Let us recall Gronwall's lemma: if $u(t)$ is a function satisfying the differential inequality

$$\frac{du}{dt} \leq \alpha(t)u(t)$$

for some function $\alpha(t)$, then we have

$$u(t) \leq u(0) \exp \int_0^t \alpha(s) ds.$$

If $T(0) \leq 1$, we can write the equation for $T - 1$ as

$$\frac{d(T - 1)}{dt} = -\gamma S(T - 1) - \delta_M M(T - 1) - \gamma S \leq (-\gamma S - \delta_M M)(T - 1)$$

Thus, by Gronwall's lemma, we have

$$T(t) - 1 \leq (T(0) - 1) \exp \int_0^t (-\gamma S(s) - \delta_M M(s)) ds \leq 0$$

which proves that $T(t) \leq 1$ for all $t \geq 0$.

In the other case, if $T(0) > 1$, we can write the equation for $T - T(0)$ as

$$\frac{d(T - T(0))}{dt} = -\gamma S(T - T(0)) - \delta_M M(T - T(0)) + \delta_M M(1 - T(0)) - \gamma ST(0) \leq (-\gamma S - \delta_M M)(T - T(0))$$

Thus, by Gronwall's lemma, we have

$$T(t) - T(0) \leq (T(0) - T(0)) \exp \int_0^t (-\gamma S(s) - \delta_M M(s)) ds = 0$$

which proves that $T(t) \leq T(0)$ for all $t \geq 0$. $\square$

About the variables S , N and M , we did not include degradation terms in the S , N and M equations, which could ensure the boundedness of S , N and M . Indeed we shall see in Section 4 that in some cases the solution may not be bounded for all time, at least for S , N and M . Actually we are interested in the immune response to an external stimulus, which can lead to a very high level of inflammation and cell recruitment from the bloodstream or marrow. The transient dynamics of the system can involve large variations in the levels of inflammation and cell populations, which may be shadowed by a model that includes degradation terms that dampens the growth of these variables. In our model, the solution may not be bounded for all time, but it can still be biologically relevant since we are interested in the early crisis dynamics rather than the long-term behavior of the system.

However, we can still derive some lower bounds for the variables M and N . Using Gronwall's lemma like in the previous proof, we can also bound from below the variables M and N .

Proposition 3.3. *Assume that the initial condition $M(0)$ (resp. $N(0)$) is greater than or equal to 1, then the variable $M(t)$ (resp. $N(t)$) is greater than or equal to 1 for all $t \geq 0$. If $M(0) < 1$ (resp. $N(0) < 1$), then $M(t)$ (resp. $N(t)$) is greater than or equal to $M(0)$ (resp. $N(0)$) for all $t \geq 0$.*

Remark 3.2. *The upper boundedness of T and the lower bounds for M and N are biologically relevant since they reflect the regulatory mechanisms that control the populations of SL-TRM, Momac, and neutrophils within the joint tissue. The population of SL-TRM cells is limited by spatial constraints and regulatory feedback, while the populations of Momac and neutrophils cannot fall below certain levels due to their large supply in blood and marrow.*

is it biologically correct?

3.3 Equilibria points

Equilibria points of the system are found by setting the right-hand side of the system to zero, i.e. the equilibrium points (S^*, N^*, M^*, T^*) satisfy the following system of algebraic equations:

$$(aN^* - bT^*)S^* = 0 \quad (2)$$

$$\alpha S^* N^* - \delta_N N^* (N^* - 1) = 0 \quad (3)$$

$$\beta S^* - \delta_M M^* (M^* - 1) = 0 \quad (4)$$

$$-\gamma S^* T^* + \delta_M M^* (1 - T^*) = 0 \quad (5)$$

3.3.1 Inflammation free equilibria ($S^* = 0$).

A first natural equilibrium point is given by $S^* = 0, N^* = 1, M^* = 1, T^* = 1$. This point corresponds to the disease-free state, where there is no inflammation ($S^* = 0$) and the populations of neutrophils, Momac, and SL-TRM are at their baseline levels ($N^* = 1, M^* = 1, T^* = 1$).

There are other equilibria points with $S^* = 0$ namely $S^* = 0, N^* = 0, M^* = 0, T^* = R$, with arbitrary R ; $S^* = 0, N^* = 0, M^* = 1, T^* = 1$; $S^* = 0, N^* = 1, M^* = 0, T^* = R$, with arbitrary R .

Remark 3.3. *These equilibria points with $N^* = 0$ or $M^* = 0$ are not biologically relevant since they correspond to the absence of neutrophils or Momac cells, which is not realistic in a living organism. Furthermore, we will prove later that these equilibria points are unstable, so they will not be observed in physiological conditions.*

3.3.2 Equilibria points with $S^* \neq 0$.

We express N^* in terms of T^* from equation (2):

$$N^* = \frac{b}{a} T^* \quad (6)$$

Substituting this expression for N^* into equation (3), we obtain:

$$S^* = \frac{\delta_N}{\alpha} \left(\frac{b}{a} T^* - 1 \right) \quad (7)$$

Since we are now looking for equilibria points with $S^* \neq 0$, (4) and (5) imply both $M^* \neq 1$ and $T^* \neq 1$. From equation (4) and equation (5), we derive $\delta_M M^* = \frac{\beta S^*}{M^* - 1}$ and $\delta_M M^* = \frac{\gamma S^* T^*}{1 - T^*}$. Equating these two expressions for $\delta_M M^*$, we obtain:

$$\frac{\beta S^*}{M^* - 1} = \frac{\gamma S^* T^*}{1 - T^*}$$

Assuming $S^* \neq 0$, we can simplify this equation to:

$$\frac{\beta}{M^* - 1} = \frac{\gamma T^*}{1 - T^*}$$

From this, we can express M^* in terms of T^* :

$$M^* = 1 + \frac{\beta(1 - T^*)}{\gamma T^*} = 1 - \frac{\beta}{\gamma} + \frac{\beta}{\gamma T^*} \quad (8)$$

Substituting the expression for M^* and S^* into the last equation (5), we obtain

$$-\gamma \frac{\delta_N}{\alpha} \left(\frac{b}{a} T^* - 1 \right) T^* + \delta_M \left(1 - \frac{\beta}{\gamma} + \frac{\beta}{\gamma T^*} \right) (1 - T^*) = 0 \quad (9)$$

Since $T^* \neq 0$, we can multiply this equation by T^* and rearrange the terms to obtain a cubic equation for T^* :

$$\frac{\gamma}{\alpha} \frac{\delta_N}{\delta_M} \frac{b}{a} (T^*)^3 + \left(1 - \frac{\gamma}{\alpha} \frac{\delta_N}{\delta_M} - \frac{\beta}{\gamma} \right) (T^*)^2 + \left(2 \frac{\beta}{\gamma} - 1 \right) T^* - \frac{\beta}{\gamma} = 0 \quad (10)$$

Rewriting this equation in the standard unitary form $(T^*)^3 + p(T^*)^2 + qT^* + r = 0$, we have:

$$(T^*)^3 + \frac{\alpha}{\gamma} \frac{\delta_M}{\delta_N} \frac{a}{b} \left(1 - \frac{\beta}{\gamma} - \frac{\gamma}{\alpha} \frac{\delta_N}{\delta_M} \right) (T^*)^2 + \frac{\alpha}{\gamma} \frac{\delta_M}{\delta_N} \frac{a}{b} \left(2 \frac{\beta}{\gamma} - 1 \right) T^* - \frac{\alpha}{\gamma} \frac{\delta_M}{\delta_N} \frac{a}{b} \frac{\beta}{\gamma} = 0 \quad (11)$$

Notice that we need only 3 *non dimensional* parameters a/b , β/γ , $\frac{\gamma \delta_N}{\alpha \delta_M}$ instead of the 7 original parameters. In the following, we denote the 3 *non dimensional* parameters as $A = a/b$, $B = \beta/\gamma$, and $C = \frac{\gamma \delta_N}{\alpha \delta_M}$.

Remark 3.4. *The parameters A , B , and C have the following biological interpretations:*

- $A = a/b$. *This parameter represents the ratio of the inflammatory effect of the neutrophils w.r.t. the anti-inflammatory effect of the macrophages.*
- $B = \beta/\gamma$. *This parameter represents the ratio of the increasing rate of the Momac to the decreasing rate of the macrophage.*
- $C = \frac{\gamma \delta_N}{\alpha \delta_M} = \left[\frac{\gamma}{\delta_M} : \frac{\alpha}{\delta_N} \right]$. *This parameter represents the ratio of the macrophages vs the neutrophiles effects in the disease progression.*

Using these parameters, the cubic equation in T^* becomes:

$$(T^*)^3 + \frac{A}{C}(1 - B - C)(T^*)^2 + \frac{A}{C}(2B - 1)T^* - \frac{AB}{C} = 0 \quad (12)$$

We now determine the number of positive roots of the cubic equation (12). From basic algebra, we can show that there are at most 3 such intersection points, hence at most 3 equilibria points with $S^* \neq 0$. We have to take into account the biological feasibility which imposes the roots to be real nonnegative numbers. Since $\frac{AB}{C} > 0$, there is at least one positive root. But we can be more precise using *Descartes' rule of signs*. It states that the number of positive real roots of a polynomial (counting multiplicities) is equal to the number of sign changes between consecutive nonzero coefficients, or less than that by an even number. In our case, the coefficients of the cubic polynomial in (12) are:

$$1, \quad \frac{A}{C}(1 - B - C), \quad \frac{A}{C}(2B - 1), \quad -\frac{AB}{C}$$

So the number of sign changes depends on the signs of the second and third coefficients. We have the following cases:

- If $B + C > 1$ and $B > 1/2$, then there are 3 sign changes and the cubic has one or three positive roots, which can collapse according to their multiplicities.
- If $B + C < 1$ and $B > 1/2$, then there is one sign change and the cubic has only one positive roots.
- If $B + C < 1$ and $B < 1/2$, then there is one sign changes and the cubic has only one positive root.
- If $B + C > 1$ and $B < 1/2$, then there is one sign change and the cubic has only one positive root.

We can plot some typical cases in the following figure (see figure 2).

We can use Cardano's method to find the roots of the cubic equation (12). We rewrite it in the standard form:

$$T^3 + pT^2 + qT + r = 0$$

where

$$p = \frac{A}{C}(1 - B - C), \quad q = \frac{A}{C}(2B - 1), \quad r = -\frac{AB}{C}$$

We perform the change of variable $T = x - p/3$ to eliminate the quadratic term and obtain the depressed cubic equation:

$$x^3 + \mathcal{A}x + \mathcal{B} = 0$$

where

$$\mathcal{A} = q - \frac{p^2}{3}, \quad \mathcal{B} = \frac{2p^3}{27} - \frac{pq}{3} + r$$

The discriminant of this equation is given by:

$$\Delta_3 = \left(\frac{\mathcal{B}}{2}\right)^2 + \left(\frac{\mathcal{A}}{3}\right)^3$$

The nature of the roots depends on the sign of the discriminant Δ_3 :

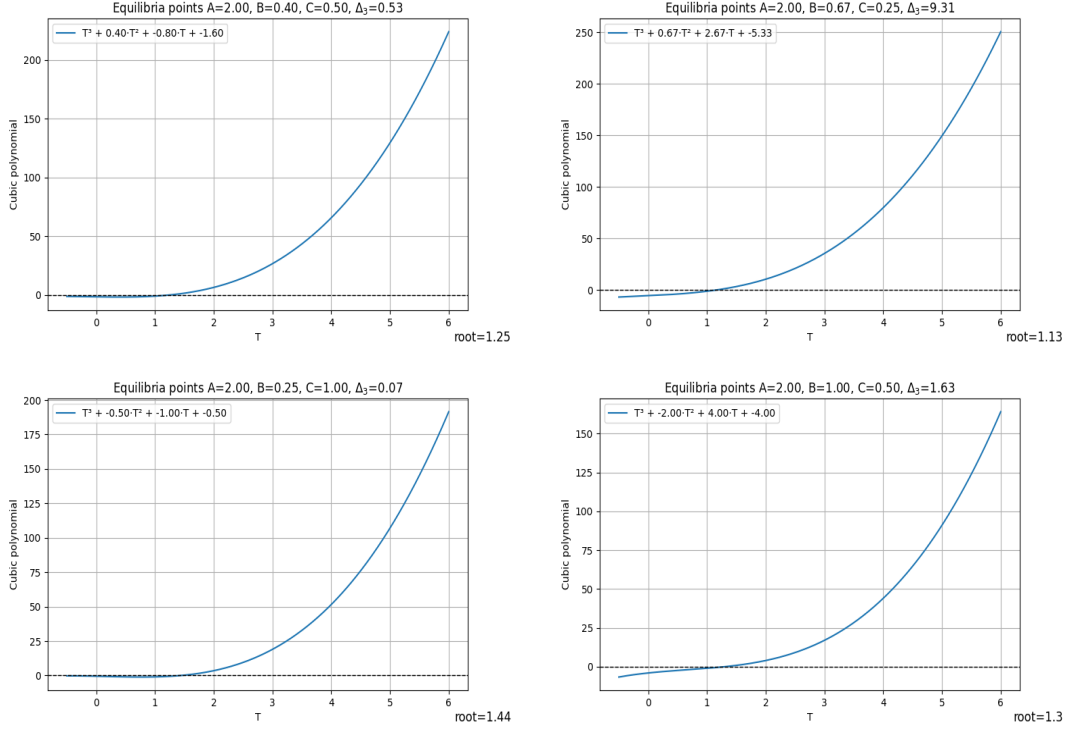


Figure 2: Cubic equation in T^* , left $B < 1/2$, right $B > 1/2$.

- If $\Delta_3 > 0$, there is one real root and two complex conjugate roots.
- If $\Delta_3 = 0$, all roots are real and at least two are equal.
- If $\Delta_3 < 0$, all roots are real and unequal.

Hence we can characterize the case when we have three distinct real roots by the condition $\Delta_3 < 0$, which is equivalent to:

$$\left(\frac{\mathcal{A}}{3}\right)^3 < -\left(\frac{\mathcal{B}}{2}\right)^2.$$

The parameters $\mathcal{A}$ and $\mathcal{B}$ can be rewritten in terms of the original parameters using computer algebra as follows:

$$\mathcal{A} = \frac{A}{3C^2} \left(-A(B+C-1)^2 + 3C(2B-1) \right)$$

$$\mathcal{B} = \frac{A}{27C^3} \left(-2A^2(B+C-1)^3 + 9AC(2B-1)(B+C-1) - 27BC^2 \right)$$

The discriminant Δ_3 can be rewritten in terms of the original parameters as follows:

$$\Delta_3 = \frac{A^2}{108C^4} \left(A^2(4BC-1)(B+C-1)^2 - 2AC(2B-1)(B^2-9BC-B-2) + 27B^2C^2 \right)$$

The real roots can be found using Cardano's formulas. If $\Delta_3 > 0$, the real root is given by:

$$x^* = \sqrt[3]{-\frac{\mathcal{B}}{2} + \sqrt{\Delta_3}} + \sqrt[3]{-\frac{\mathcal{B}}{2} - \sqrt{\Delta_3}}$$

If $\Delta_3 < 0$, the three real roots are given by:

$$x_k^* = 2\sqrt{-\frac{\mathcal{A}}{3}} \cos \left(\frac{1}{3} \arccos \left(\frac{3\mathcal{B}}{2\mathcal{A}} \sqrt{-\frac{3}{\mathcal{A}}} \right) + \frac{2k\pi}{3} \right), \quad k = 0, 1, 2$$

If $\Delta_3 = 0$, the real roots are given by:

$$x_1^* = 2\sqrt[3]{-\frac{\mathcal{B}}{2}}, \quad x_2^* = x_3^* = \sqrt[3]{\frac{\mathcal{B}}{2}}$$

Here we plot a case with three distinct positive roots (see figure 3). Namely we take $A = 1.1$, $B = 5$ and $C = \frac{2}{3}$.

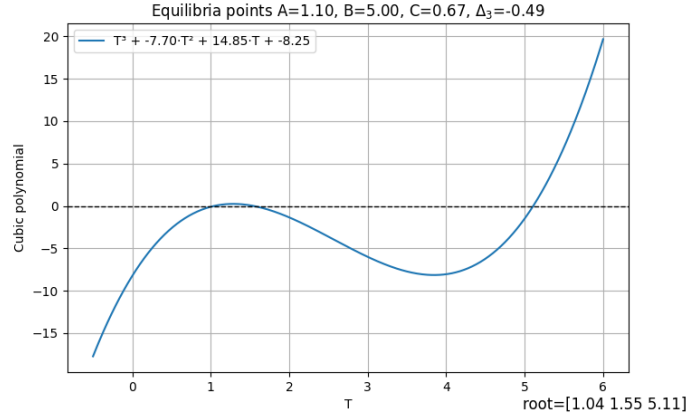


Figure 3: Cubic equation with three distinct real roots.

Finally, we can recover the roots T^* of the original cubic equation using the relation $T^* = x^* - p/3$.

We have to make sure that the nonnegative roots T^* found are biologically feasible, i.e. that they lead to nonnegative values of S^* , N^* , and M^* . For instance in the case $A = 1.1$, $B = 5$, $C = 2/3$ plotted in figure 3, we have three distinct positive roots of the cubic equation (12), but none satisfying the biological feasibility conditions.

From equations (6), (7), and (8), we have:

$$N^* = \frac{b}{a}T^*, \quad S^* = \frac{\delta_N}{\alpha} \left(\frac{b}{a}T^* - 1 \right), \quad M^* = 1 - \frac{\beta}{\gamma} + \frac{\beta}{\gamma T^*}$$

Thus, the conditions for biological feasibility are:

- $S^* \geq 0$ iff $T^* \geq \frac{a}{b} = A$
- $N^* \geq 0$ iff $T^* \geq 0$ (always true since we consider only positive roots)
- $M^* \geq 0$ iff $T^* \leq \frac{\beta}{\beta-\gamma} = \frac{B}{B-1} = 1 + \frac{1}{B-1}$ if $B = \frac{\beta}{\gamma} > 1$; always true if $B \leq 1$.

So the biological feasibility conditions can be summarized as follows:

$$T^* \geq A \tag{13}$$

$$T^* \leq 1 + \frac{1}{B-1} \quad \text{if } B > 1 \tag{14}$$

Let us consider first the particular case $A = a/b = 1$. In this case, we have an obvious solution $T^* = 1$ which corresponds to the disease-free equilibrium point. Therefore we can factor the cubic polynomial as follows:

$$(T^* - 1) \left((T^*)^2 - \frac{1}{C}(B-1)T^* + \frac{B}{C} \right) = 0. \quad (15)$$

If $B \leq 1$, all the coefficient of the quadratic polynomial are positive, so there is only one equilibrium, the disease-free $(S^*, N^*, M^*, T^*) = (0, 1, 1, 1)$. Otherwise if $B > 1$ the feasibility conditions (13)-(14) reduce to:

$$1 \leq T^* \leq 1 + \frac{1}{B-1}. \quad (16)$$

We can check easily that only the disease-free root $T^* = 1$ satisfies this condition, while the other two roots are outside this interval, as we prove now. Indeed if the discriminant of the quadratic factor $\Delta = \frac{(B-1)^2 - 4BC}{C^2} \geq 0$, we have $C \leq \frac{(B-1)^2}{4B}$, hence $\frac{B-1}{2C} \geq \frac{B-1}{2 \frac{(B-1)^2}{4B}} = \frac{2B}{B-1} \geq 2$ so the root $T^* = \frac{B-1+\sqrt{\Delta}}{2C} \geq 2$ is not feasible. Moreover, the other root $T^* = \frac{B-1-\sqrt{\Delta}}{2C} \leq \frac{B-1}{2C}$. In order to be feasible, this root should satisfy $\frac{B-1-\sqrt{\Delta}}{2C} \leq \frac{B}{B-1}$, which implies $C \geq \frac{(B-1)^2}{2B}$, contradicting the assumption $\Delta \geq 0$.

In the general case $A \neq 1$, the feasibility conditions (13)-(14) have to be checked for each positive root found. For $A > 1$, we will show that none of the positive roots satisfy the feasibility conditions (see figure 3) whereas for $A < 1$, there is always exactly one feasible root (in addition to the disease-free equilibrium). To this aim, we consider four cases depending on the signs of $A - 1$ and $B - 1$.

- Case $A > 1$ and $B \geq 1$.

Let us define $f(T^*)$ as the cubic polynomial in the left-hand side of equation (12).

If $B > 1$ we have

$$\begin{aligned} f(T^*) &= T^{*3} - \frac{A}{C}(B-1)T^{*2} - AT^{*2} + \frac{A}{C}(B-1)T^* + \frac{A}{C}BT^* - \frac{AB}{C} \\ &= \frac{A}{C}(B-1)\left(T^* - \frac{B}{B-1}\right)(1 - T^*) + T^{*2}(T^* - A) \end{aligned}$$

So $f(T^*) > 0$, for $T^* \in [A, \frac{B}{B-1}]$ if $B > 1$.

If $B = 1$ then $f(T^*) = T^{*2}(T^* - A) + \frac{A}{C}(T^* - 1) > 0$ for $T^* > A$.

Hence, thanks to (13) and (14), there is no feasible root in this case.

- Case $A > 1$ and $B < 1$. Since

$$f(T^*) = \frac{A}{C}(B-1)\left(T^* - \frac{B}{B-1}\right)(1 - T^*) + T^{*2}(T^* - A) > 0$$

for $T^* \geq A$ and $B < 1$, there is no feasible root in this case either.

- Case $A < 1$ and $B \geq 1$.

We compute $f(1) = 1 - A > 0$ and $f(A) = (A-1)\frac{A}{C}(A(1-B) + B) < 0$, so there is at least one feasible root in this case.

If $B > 1$, computing the derivative $f'(T^*)$ we have:

$$f'(T^*) = \frac{A}{C}(B-1)\left[\left(\frac{B}{B-1} - T^*\right) + (1 - T^*)\right] + 2T^*(T^* - A) + T^{*2} > 0$$

for $T^* \in [A, 1]$. So there is exactly one root in the interval $[A, 1]$. Since once again $f(T^*) = \frac{A}{C}(B-1)(T^* - \frac{B}{B-1})(1-T^*) + T^{*2}(T^* - A)$ is positive for $T^* \in [1, \frac{B}{B-1}]$, there is exactly one feasible root in this case.

If $B = 1$, we have $f(T^*) = T^{*2}(T^* - A) + \frac{A}{C}(T^* - 1) > 0$ for $T^* > 1$ and $f'(T^*) = T^{*2} + 2T^*(T^* - A) + \frac{A}{C} > 0$ for $T^* \in [A, 1]$. So there is exactly one feasible root in this case too.

- Case $A < 1$ and $B < 1$.

Since $f(1) = 1 - A > 0$ and $f(A) = (A-1)\frac{A}{C}(A(1-B) + B) < 0$, there is at least one feasible root in this case too. Let $\tilde{T} = T^* - A$ and $\tilde{f}(\tilde{T}) = f(\tilde{T} + A)$. Then the equation $\tilde{f}(\tilde{T}) = 0$ is written as:

$$\begin{aligned} \tilde{T}^3 + \left(3A + \frac{A}{C}(1-B-C)\right)\tilde{T}^2 + \left(3A^2 + 2\frac{A^2}{C}(1-B-C) + \frac{A}{C}(2B-1)\right)\tilde{T} \\ + \frac{A}{C}(A-1)(A(1-B) + B) \\ = \tilde{T}^3 + \left(2A + \frac{A}{C}(1-B)\right)\tilde{T}^2 + \left(A^2 + 2\frac{A^2}{C}(1-B) + \frac{A}{C}(2B-1)\right)\tilde{T} \\ + \frac{A}{C}(A-1)(A(1-B) + B) = 0. \end{aligned}$$

Using *Descartes' rule of signs* we can see that there is exactly one positive root $\tilde{T}$ of the cubic equation $\tilde{f}(\tilde{T}) = 0$ as the coefficients of $\tilde{T}^3$ and $\tilde{T}^2$ are positive and the constant term is negative. So there is exactly one feasible root $T^* \geq A$ in this case too.

Let us summarize. The feasible equilibria points of the system can be classified as follows:

- Disease-free equilibrium: $(S^*, N^*, M^*, T^*) = (0, 1, 1, 1)$.
- Other equilibria with $S^* = 0$: $S^* = 0, N^* = 0, M^* = 0, T^* = R$, with arbitrary R ; $S^* = 0, N^* = 0, M^* = 1, T^* = 1$; $S^* = 0, N^* = 1, M^* = 0, T^* = R$, with arbitrary R .
- Feasible equilibria with $S^* \neq 0$: None when $A \geq 1$. We shall see in the numerical simulations that in this case the system quickly diverges away from the disease-free equilibrium, no matter how weak is the initial inflammation. This suggests that the system is in a persistent inflammatory state with $S^* \neq 0$. When $A < 1$, there is only one biologically feasible equilibria with $S^* \neq 0$. We shall see in the next section that this equilibrium is unstable, and the numerical simulations suggest that the system converges to the disease-free equilibrium, hence the inflammation is resolved.

3.4 Stability analysis

Linear stability analysis of the equilibria point (S^*, N^*, M^*, T^*) is done by linearizing the system around this point. The Jacobian matrix of the system is given by:

$$J = \begin{pmatrix} aN - bT & aS & 0 & -bS \\ \alpha N & \alpha S - \delta_N(2N - 1) & 0 & 0 \\ \beta & 0 & -\delta_M(2M - 1) & 0 \\ -\gamma T & 0 & \delta_M(1 - T) & -\gamma S - \delta_M M \end{pmatrix}$$

According to the linear stability theory, the equilibrium point is (locally asymptotically) stable if all eigenvalues of the Jacobian matrix have negative real parts, and unstable if at least one eigenvalue has a positive real part. In what follows, we analyze the stability of the different equilibria points found in the previous section. By locally asymptotically stable, we mean that if the system is perturbed slightly from the equilibrium point, it will return to the equilibrium point as time goes to infinity. On the other hand, if the equilibrium point is unstable, a small perturbation will cause the system to move away from the equilibrium point.

3.4.1 Stability of inflammation free equilibria ($S^* = 0$).

We analyze first the stability of the disease-free equilibrium point $(S^*, N^*, M^*, T^*) = (0, 1, 1, 1)$. At this point, the Jacobian matrix simplifies to:

$$J|_{(S^*, N^*, M^*, T^*)=(0,1,1,1)} = \begin{pmatrix} a-b & 0 & 0 & 0 \\ \alpha & -\delta_N & 0 & 0 \\ \beta & 0 & -\delta_M & 0 \\ -\gamma & 0 & 0 & -\delta_M \end{pmatrix}$$

We notice immediately that the eigenvalues of this matrix are $a-b$, $-\delta_N$ and $-\delta_M$. The last two are negative since $\delta_N, \delta_M > 0$. The remaining eigenvalue is $a-b$. Thus the disease-free equilibrium is stable if $a < b$ and unstable if $a > b$. This means that if the inflammatory effect of Neutrophils cells controlled by parameter a is less than the curing effect of the SL-TRM macrophages controlled by parameter b , the system will return to the disease-free state. Conversely, if $a > b$, the system may move away from this equilibrium, potentially leading to a persistent inflammatory state. The particular case $a = b$ corresponds to a bifurcation point where the stability of the disease-free equilibrium changes. In this case, a more detailed analysis is needed to determine the behavior of the system near this point, which involve higher-order terms in the Taylor expansion of the system around the equilibrium. The numerical simulations, see section 4 suggest that for $a = b$ the system quickly diverges from the disease-free equilibrium, and indeed this equilibrium is unstable in this case as we prove below using the center manifold theory.

For $a = b$ the disease-free equilibrium is a non-hyperbolic equilibrium with eigenvalues 0 , $-\delta_N$, $-\delta_M$ and $-\delta_M$. The stability of this equilibrium can be analyzed using center manifold theory. We prove here that the center manifold is repulsive for $S^* \geq 0$, hence the equilibrium is unstable.

In this case there is one center direction and three stable directions. We define new variables $x = S$, $y = N - 1$, $z = M - 1$, and $w = T - 1$ to shift the equilibrium point to the origin. The system can be rewritten as:

$$\begin{aligned} \dot{x} &= ax(y-w) \\ \dot{y} &= (y+1)(\alpha x - \delta_N y) \\ \dot{z} &= \beta x - \delta_M z(z+1) \\ \dot{w} &= -\gamma x(w+1) - \delta_M w(z+1) \end{aligned}$$

The linear part of the system is given by:

$$\begin{aligned}\dot{x} &= 0 \\ \dot{y} &= \alpha x - \delta_N y \\ \dot{z} &= \beta x - \delta_M z \\ \dot{w} &= -\gamma x - \delta_M w\end{aligned}$$

where the center direction is given by the variable x and the stable directions are given by the variables y , z , and w . The center manifold can be expressed as a graph of the form $h(x) = (h_1(x), h_2(x), h_3(x))$ where $h_i(0) = 0$ for $i = 1, 2, 3$. We can compute the Taylor expansion of h and find that the first nonzero term is given by $h_1(x) = \frac{\alpha}{\delta_N}x + o(x)$, $h_2(x) = \frac{\beta}{\delta_M}x + o(x)$, and $h_3(x) = -\frac{\gamma}{\delta_M}x + o(x)$. The dynamics on the center manifold is given by:

$$\dot{x} = ax^2\left(\frac{\alpha}{\delta_N} + \frac{\gamma}{\delta_M}\right) + o(x^2)$$

Since $\frac{\alpha}{\delta_N} + \frac{\gamma}{\delta_M} > 0$, the center manifold is repulsive for $x \geq 0$, hence the equilibrium is unstable.

The stability of other equilibria points can be analyzed similarly by evaluating the Jacobian matrix at those points and determining the eigenvalues.

For $S^* = 0, N^* = 0, M^* = 0, T^* = R$, with arbitrary R , the Jacobian matrix becomes:

$$J|_{(S^*, N^*, M^*, T^*)=(0,0,0,R)} = \begin{pmatrix} -bR & 0 & 0 & 0 \\ 0 & \delta_N & 0 & 0 \\ \beta & 0 & \delta_M & 0 \\ -\gamma R & 0 & \delta_M(1-R) & 0 \end{pmatrix}$$

The eigenvalues of this matrix are $-bR$, δ_N , δ_M , and 0. Since δ_N and δ_M are positive, this equilibrium point is unstable. Even if the immune signal will decrease exponentially fast, the populations of neutrophils and Momac will grow out of bound. This unstable equilibrium where the inflammation is zero as well as the population of neutrophils and Momac is highly unlikely to be observed in practice.

For $S^* = 0, N^* = 0, M^* = 1, T^* = 1$, the Jacobian matrix is:

$$J|_{(S^*, N^*, M^*, T^*)=(0,0,1,1)} = \begin{pmatrix} -b & 0 & 0 & 0 \\ 0 & \delta_N & 0 & 0 \\ \beta & 0 & -\delta_M & 0 \\ -\gamma & 0 & 0 & -\delta_M \end{pmatrix}$$

The eigenvalues of this matrix are $-b$, δ_N , $-\delta_M$, and $-\gamma$. Since there is a positive eigenvalue, this equilibrium point is unstable. Neutrophils can grow out of bound. This equilibrium will not be observed in practice neither.

For $S^* = 0, N^* = 1, M^* = 0, T^* = R$, with arbitrary R , the Jacobian matrix is

$$J|_{(S^*, N^*, M^*, T^*)=(0,1,0,R)} = \begin{pmatrix} a - bR & 0 & 0 & 0 \\ \alpha & -\delta_N & 0 & 0 \\ \beta & 0 & \delta_M & 0 \\ -\gamma R & 0 & \delta_M(1-R) & 0 \end{pmatrix}$$

There is a positive eigenvalue δ_M , hence this equilibrium point is unstable. Momac can grow out of bound. This equilibrium will not be observed in practice neither.

3.4.2 Stability of equilibrium with $S^* \neq 0$.

The stability of the equilibria with $S^* \neq 0$ can be analyzed similarly by evaluating the Jacobian matrix at those points and determining the eigenvalues. Taking into account the equilibrium equations (2), (3), and (5), the Jacobian matrix at (S^*, N^*, M^*, T^*) is:

$$J = \begin{pmatrix} 0 & aS & 0 & -bS \\ \alpha N & -\delta_N(N) & 0 & 0 \\ \beta & 0 & -\delta_M(2M-1) & 0 \\ -\gamma T & 0 & \delta_M(1-T) & -\delta_M \frac{M}{T} \end{pmatrix}$$

The eigenvalues of this matrix can be found by solving the characteristic polynomial $\det(J - \lambda I) = 0$. A straightforward computation shows that the characteristic polynomial is given by:

$$\begin{aligned} P(\lambda) = & \lambda^4 + \left(\delta_N N + \delta_M(2M-1) + \delta_M \frac{M}{T} \right) \lambda^3 \\ & + \left(\delta_N N \delta_M(2M-1 + \frac{M}{T}) + \delta_M^2(2M-1) \frac{M}{T} - aS\alpha N - bS\beta\gamma \right) \lambda^2 \\ & + \left(-aS\alpha N \delta_M(2M-1 + \frac{M}{T}) - bS\gamma T \delta_N N + \delta_N N \delta_M^2(2M-1) \frac{M}{T} - b\delta_M^2 M^2(1-T) \right) \lambda \\ & - aS\alpha N \delta_M^2(2M-1) \frac{M}{T} - b\delta_N N \delta_M^2 M^2(1-T) \end{aligned}$$

Due to the Descartes' rule of signs, we claim that the polynomial $P(\lambda)$ has one positive root (hence the equilibrium is unstable). Indeed, the coefficients of $P(\lambda)$ are positive for the terms in λ^4 and λ^3 , whereas the constant term is negative. To prove that there is only one change of sign and hence one positive root, we have to check that if the coefficient of λ is positive then the coefficient of λ^2 is positive too.

If the coefficient of λ is positive we have the following inequality:

$$aS\alpha N \delta_M(2M-1 + \frac{M}{T}) + b\delta_M^2 M^2(1-T) + bS\gamma T \delta_N N < \delta_N N \delta_M^2(2M-1) \frac{M}{T}$$

Thanks to the positivity of all terms, we can deduce the following inequalities:

$$aS\alpha N \delta_M(2M-1 + \frac{M}{T}) < \delta_N N \delta_M^2(2M-1) \frac{M}{T}, \quad (17)$$

and

$$bS\gamma T < \delta_M^2(2M-1) \frac{M}{T}. \quad (18)$$

Inequality (17) implies that

$$aS\alpha N < aS\alpha N \left(\frac{\delta_M^2(2M-1 + \frac{M}{T})^2}{\delta_M^2(2M-1) \frac{M}{T}} \right) < \delta_N N \delta_M(2M-1 + \frac{M}{T}). \quad (19)$$

Due to (18) and (19), the coefficient of λ^2 is positive.

We proved that biologically feasible equilibrium with $S^* \neq 0$ only exists if $A < 1$ and is unstable. For $A \geq 1$ the only biologically feasible equilibrium is the disease-free equilibrium which is unstable, as we have seen.

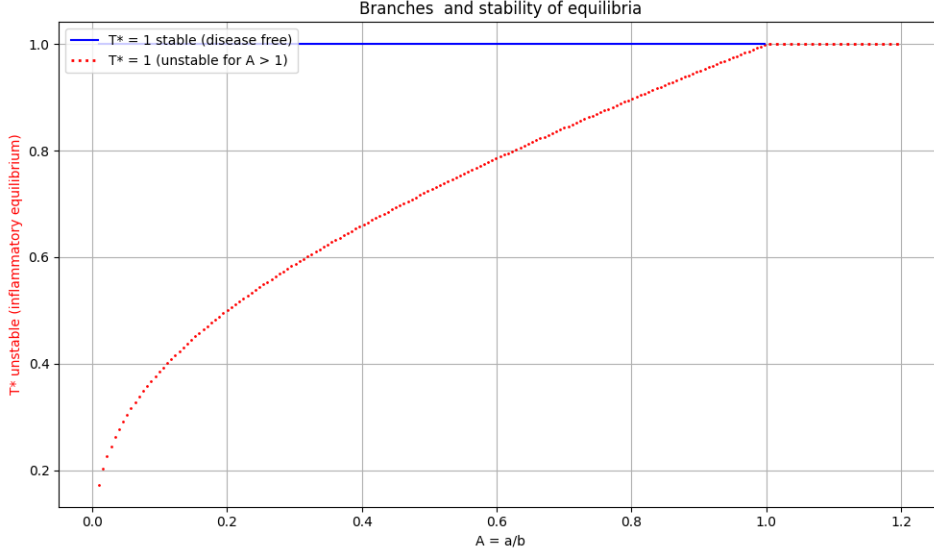


Figure 4: Bifurcation diagram in the (A, T^*) plane.

3.5 Bifurcation diagram

We perform a bifurcation analysis with respect to the *non dimensional* parameters $A = a/b$. We restrict to the biologically relevant feasible equilibria. From the stability analysis above, we see that we have only one bifurcation at $A = 1$: for $A < 1$ there are two biologically feasible equilibria, one inflammatory one with $S^* \neq 0$ which is unstable and the disease-free equilibrium which is stable; for $A \geq 1$ the disease-free equilibrium is unstable and there is no other biologically feasible equilibrium. The number and stability of equilibria change as the parameter A crosses the critical value $A = 1$, indicating bifurcation points in the system's dynamics. We can plot the bifurcation diagram in the (A, S^*) plane (see figure 4).

4 Numerical simulations

Here we reproduce some typical simulations of the model using the following parameters: $\alpha = 1, \beta = 1, \gamma = 1, \delta_N = 1, \delta_M = 1, a = 0.35, b = 1$, duration=20 days and initial conditions $S(0) = 1, N(0) = 1, M(0) = 1, T(0) = 1$. Now let us take $a = 0.38$, keeping the other parameters unchanged. We see clearly in fig. 5 that for $A = 0.35 < 1$ the system converges to the disease-free equilibrium, whereas for $A = 0.38$ the system diverges away from this equilibrium, indicating a chronic inflammatory state, see fig. 6. We can also decrease the initial immune signal $S(0)$ to see if the system converges to the disease-free equilibrium or diverges away from it. Halving the initial immune signal i.e. taking $S(0) = 0.5$ and $A = 0.38$, we obtain the following result: We see in fig. 7 that even if $A = 0.38$, the system converges to the disease-free equilibrium, indicating that the initial immune signal is not strong enough to trigger a persistent inflammatory state. We can also start from a depressed immunitary state, taking $S(0) = 0.5, N(0) = 1, M(0) = 0.1, T(0) = 0.05$ and $A = 0.38$. We observe in fig. 8 that the system diverges away from the disease-free equilibrium, indicating a persistent inflammatory state, even if the initial immune signal is low. We can also start from a very high initial immune signal, taking $S(0) = 5, N(0) = 1, M(0) = 1, T(0) = 1$ and $A = 0.072$. We observe in fig. 9

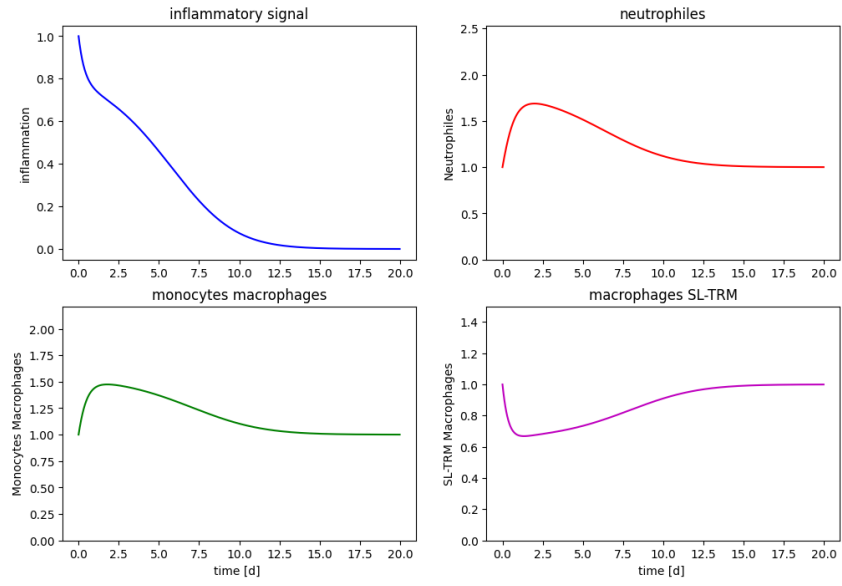


Figure 5: $A=0.35$

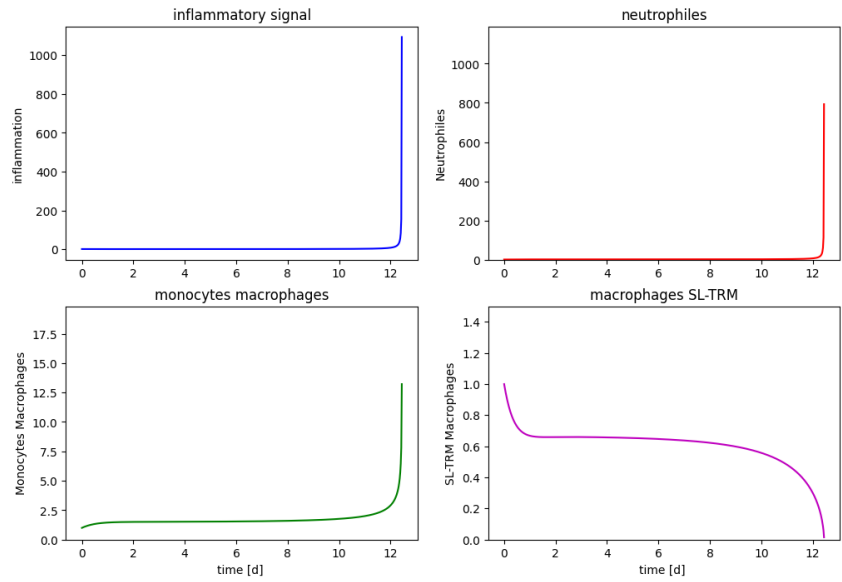


Figure 6: $A=0.38$

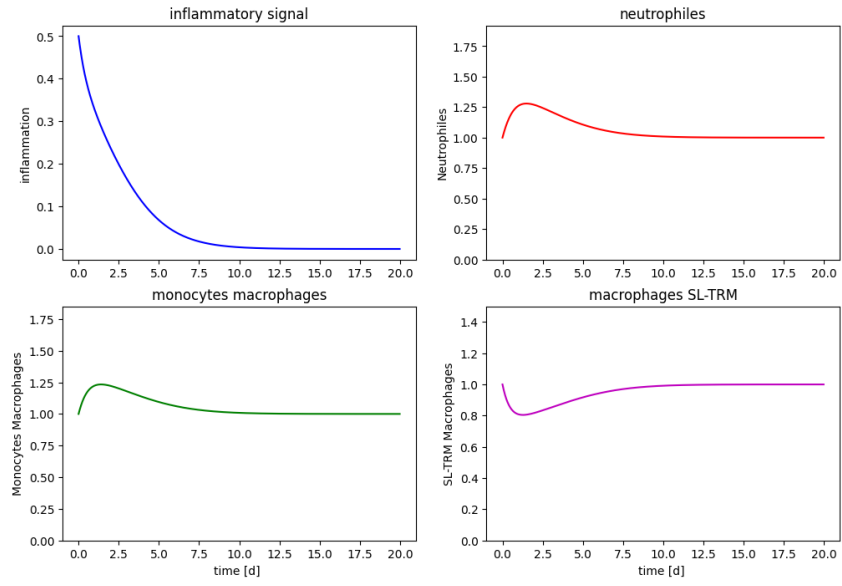


Figure 7: $A=0.38$, $S(0)=0.5$

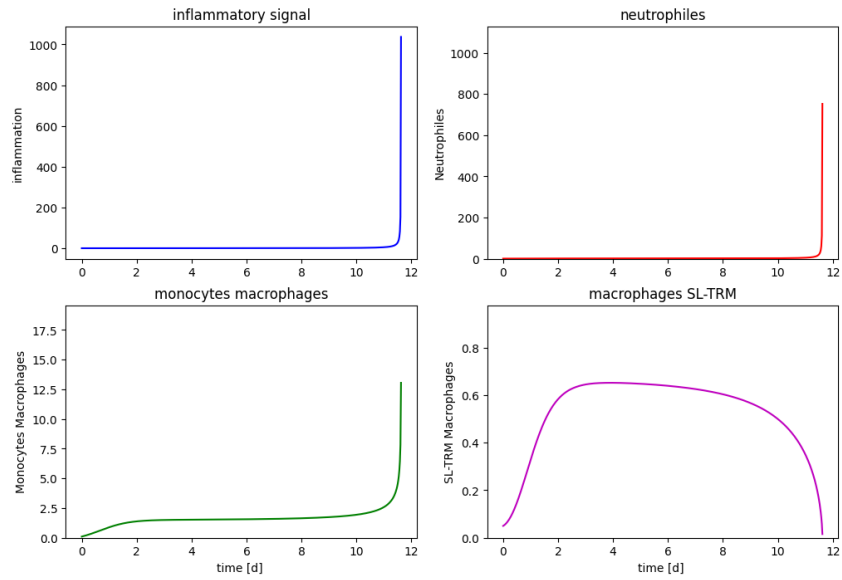


Figure 8: $A=0.38$, depressed initial conditions

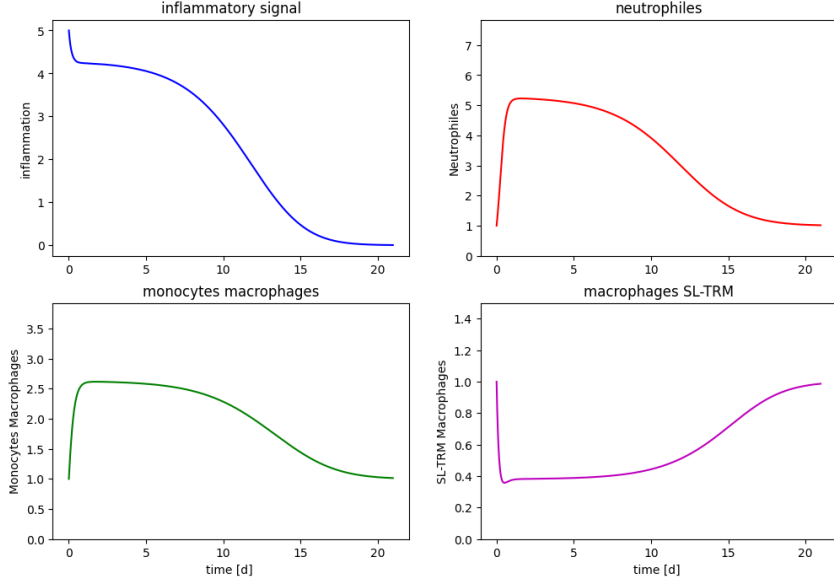


Figure 9: $A=0.072$, $S(0)=5$

that the system returns to the disease-free equilibrium, indicating that even if the initial immune signal is high, the system can still heal when $A = 0.072 < 1$. But if we slightly increase A to $A = 0.073$, keeping the other parameters unchanged, we observe in fig. 10 that the system diverges away from the disease-free equilibrium, indicating a persistent inflammatory state. To see how the parameter $A = a/b$ controls the healing prognosis

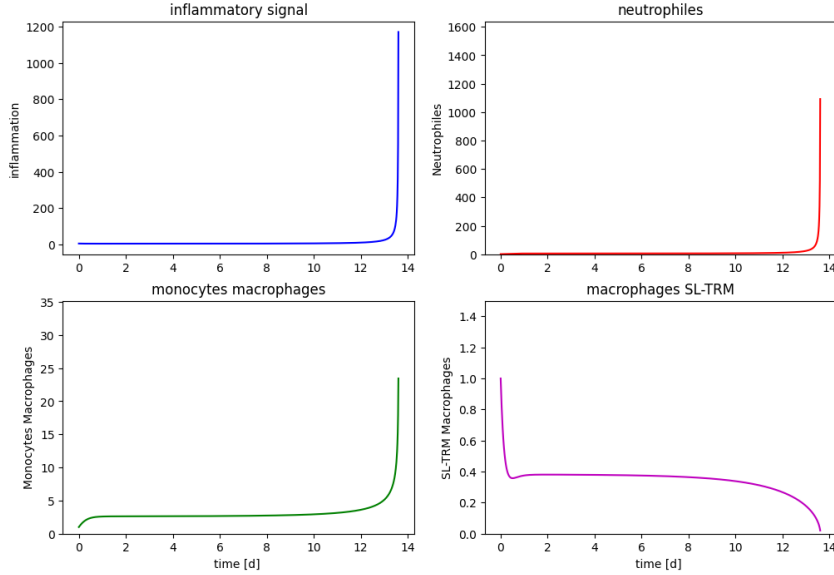


Figure 10: $A=0.073$, $S(0)=5$

we plot in Fig. 11 a (A, S) plane figure where for each value of A and starting value ($S(0) = S, N = 1, M = 1, T = 1$) we run the model for three weeks and see how far from the disease-free state the systems ends up. We color in green points where (S, M, N, T) is close to $(0, 1, 1, 1)$ up to a given tolerance (say 5%). We color in red points which end up far away from the equilibrium. We notice that for $A < 1$, the system converges to the disease-free equilibrium provided that the initial immune signal $S(0)$ is below a given

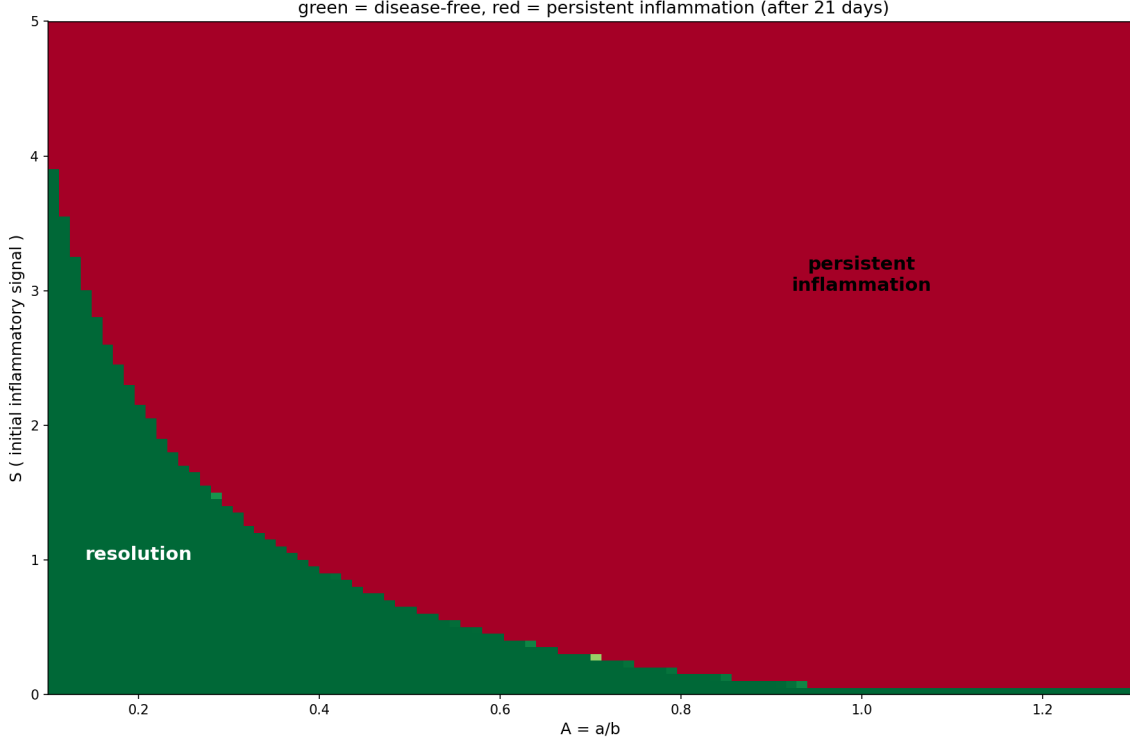


Figure 11: Healing w.r.t. $A = a/b$ and initial dose of immune signal S

threshold. This threshold increases as A decreases, meaning that the system can tolerate higher initial immune signals without diverging away from the disease-free equilibrium when A is smaller. For $S(0) > 5$, meaning the immune signal dose is five times the standard value, the system diverges away from the disease-free equilibrium for all values of A , indicating a persistent inflammatory state regardless of the value of $A > 0$ when the initial immune signal is too high. For $A > 1$, the system diverges away from the disease-free equilibrium for all values of the initial immune signal $S(0)$, indicating a persistent inflammatory state regardless of the initial conditions. This suggests that for $A > 1$ the system is in a persistent inflammatory state, where the inflammation is self-sustained and does not resolve, even if the initial immune signal is low.

The Python Jupyter notebook used to perform the simulations is available at <https://github.com/azerad/immunomath>.

5 Concluding remarks and perspectives.

In this work, we designed a simple model able to describe essential features of the monocyte-macrophages, neutrophils and sub lining tissue resident macrophages dynamics submitted to an inflammatory stimulus. We identify a single parameter controlling the positive or negative feedbacks processes at stake in arthritis. Essentially it is the ratio between the inflammatory effect of neutrophils vs the anti-inflammatory effect of SL-TRM macrophages. It suggests that drugs acting *specifically* on neutrophils and SL-TRM macrophages could be effective in curing arthritis.

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